CSMPRO00000003 Bay Area Bridge Resilience Report Tailoring

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 **Category:** Infrastructure / Bay Area

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Revision Drivers (all V2.0 documents)

1. **2025-2026 experimental and regulatory data** supersedes V1.0 specifications
2. **Solar Cycle 25 SSN ~137 peak** (observed) — elevated urgency for all Carrington-focused work
3. **Standards updates** — AASHTO 2024, IMO 2025, IEEE P2945 draft, DNV GL EMP-2, ClassNK FRP

V1.0 vs V2.0 Parameter Changes

Parameter	V1.0	V2.0	Consequence if Not Updated
Seawater conductivity	sigma=4.0 S/m	sigma=5.5 S/m (2025 NOAA Pacific Coast data)	37.5% higher GIC coupling in Bay water than V1.0 assumed
Caltrans BFRP pilot	Concept proposed in V1.0	Confirmed 2025: 72% GIC reduction; $180M program	V1.0 was speculative; V2.0 confirmed with empirical data
AB 747 compliance	Not in V1.0	CA 2024: mandatory infrastructure resilience planning	V1.0 proposal non-compliant with 2024 California law
Compound threat analysis	Not in V1.0	Seismic + GIC simultaneous scenario formally analyzed	Most dangerous scenario missed in V1.0
Upgrade to full BFRP deck	72% reduction BFRP grid	100% elimination via full structural BFRP deck	V1.0 partial solution still leaves 28% GIC exposure

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Unchanged Elements

All core design philosophy, threat physics (GIC induction equations), material selection rationale, and fundamental architecture are carried forward unchanged from V1.0. Only the parameters listed above were revised.

End Versioning Delta | CSMPRO00000003 Bay Area Bridge Resilience Report Tailoring | June 2026